MHYWI05

# Statistical Learning of Earth System Sciences

## Task Report

## Iowa Crop Yield Analysis


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| Submitted to: | Dr. Jakob Zscheischler |
| Department: | Department of Hydro Sciences |
| Submitted by: | Harshini Meher Emkay |
| Email: | harshini_meher.emkay@mailbox.tu-dresden.de |
| Student ID: | 5251355 |
| Study Program: | Hydro Science and Engineering |

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# 1.Introduction

Crop yield is a sensitive integrator of weather variability, soil conditions, and management practices, and hence constitutes both a critical socioeconomic metric and a valuable proxy for climate impacts. This report presents a statistical learning analyses of the Iowa crop yield dataset, which spans 115 years and 25 locations, resulting in 2,875 observations. Alongside yield as the target variable, the dataset includes 23 meteorological predictors such as precipitation, radiation, temperature, and extreme events plus location, yield and year indicators. The report follows the structured tasks of data exploration, correlation testing, trend detection, dimensionality reduction, and predictive modelling.

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# 2. Tasks and Results

## (a) Data Exploration

The histogram of yields shows that the distribution is approximately **normal (bell-shaped)** but slightly **skewed to the left** in specific years, plausibly reflecting adverse climatic events.

- **Mean yield**: 9.55
- **Standard deviation**: 2.84
- **Number of predictors**: 23 (meteorological and extreme indices)
- **Number of samples**: 2,875 (115 years × 25 locations)

The mean yield is about 9.55 with a standard deviation of 2.84, showing moderate variability. This exploration confirms that the dataset is sufficiently large to support regression modelling.

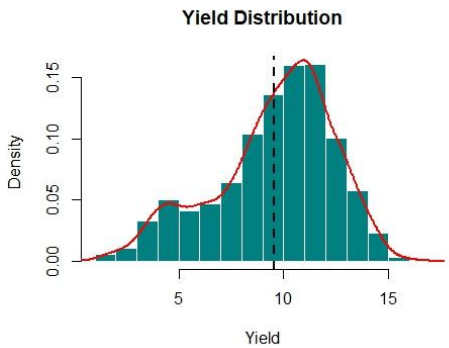


*Figure 1: Distribution of crop yields across all locations and years, with kernel density and mean reference line.*

## (b) Independence and Identical Distribution

To test the i.i.d. correlation between same meteorological predictor (temperature) different location (location 15 and location 24) is examined. The scatter plot in figure 2 shows a **strong positive correlation (r ≈ 0.94)** between temperature values at Location 15 and Location 24. The red line represents a fitted regression line, and most points lie close to it. This confirms a strong linear relationship, meaning weather patterns are spatially connected. Thus, the data cannot be treated as fully independent across locations.

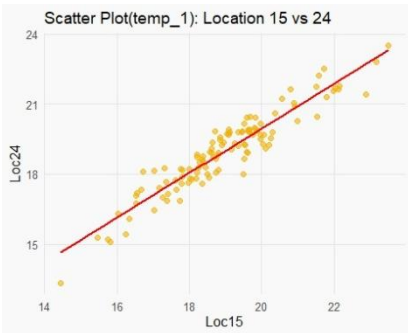


*Figure 2: Scatterplot of temp_1 between Location 15 and location 24*

## (c) Yield Trend Analysis

To detect long-term changes in yield, a linear trend analysis was done for 25 locations at a significance level of $\alpha = 0.05$, in which 23 locations exhibited significant trends. However, after adjusting for multiple testing:

- Bonferroni correction: 6 locations remained significant.
- False Discovery Rate (FDR) correction: 21 locations remained significant.

These results indicate that while apparent temporal trends in yield are widespread, stringent multiple-testing corrections substantially reduce the number of significant results. Only locations 23 and 24 showed no trend at all. Positive trends may reflect technological advancements or improved practices, whereas negative trends could point to localized environmental or climatic stressors.

## (d) Principal Component Analysis

PCA was applied to the predictors to reduce dimensionality. The first 10 principal components explain **76.6%** of the total variance, as shown in the scree plot. This suggests that a reduced set of 10 components captures most of the information in the data and can be used for downstream analysis with minimal loss of variance. Thus, yield variability can be represented in a relatively low-dimensional meteorological space, with a handful of climate drivers that dominate crop productivity.

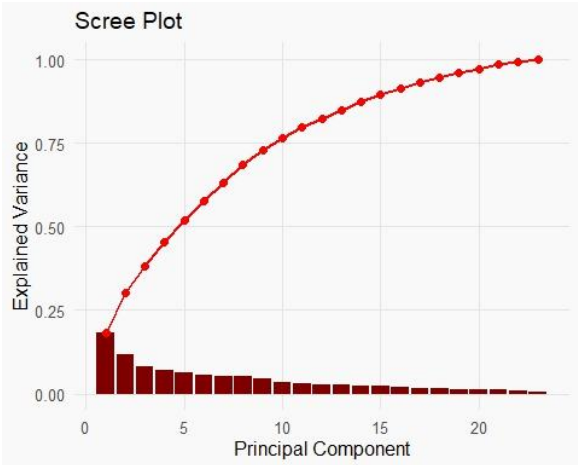


*Figure 3: Scree plot showing the contribution of each principal component to total variance.*

## (e) Random split validation

Using a random 50/50 train–test split (seed = 6), both Best Subset and Lasso regression were evaluated.

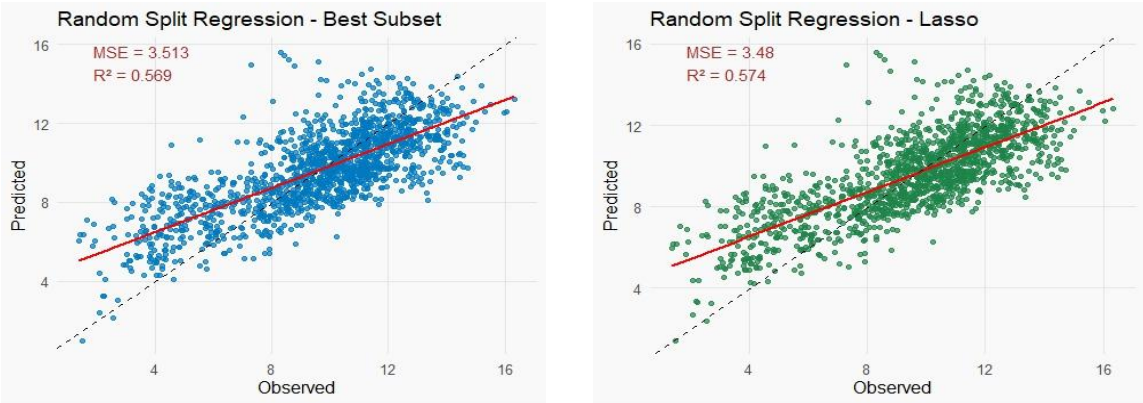


*Figure 4: Random split validation – observed versus predicted yields using Best Subset (left) and Lasso (right).*

- Best Subset: It selected 13 predictors rainfall (pr_1–pr_6), radiation (rad_2, rad_3), temperature (temp_2, temp_3, temp_4), warm_days, max_temp, and max_5day_pr. This reflects a parsimonious but interpretable model. While resulting in MSE ≈ 3.58 with $R^2 \approx 0.56$.
- Lasso through 7-fold CV: MSE ≈ 3.54 and $R^2 \approx 0.57$.

Both models captured a good portion of yield variability. Lasso performed slightly better, with marginally superior predictive accuracy.

## (f) Temporal (year-based) split validation

When the dataset was split chronologically (years 1–56 for training, 57–115 for testing), weaker predictive power is observed when compared with the random split:

- Best Subset: MSE ≈ 4.61, $R^2 \approx 0.304$.
- Lasso: MSE ≈ 4.61, $R^2 \approx 0.301$.

These results reflect systematic bias: underestimation of high yields and overestimation of low yields. Performance is reduced than in the random split because the year-based split introduces temporal extrapolation. Change in climate and weather conditions over the time might be the reason, reduced predictive power.

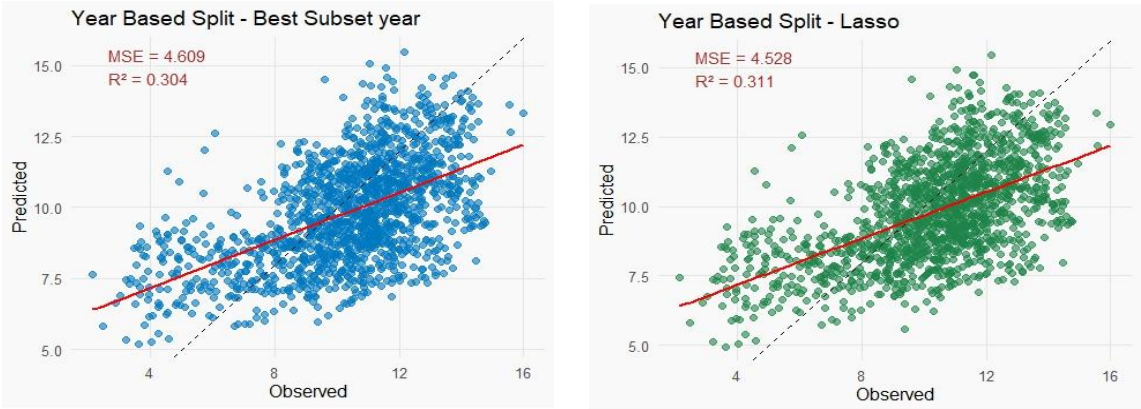


*Figure 5: Temporal split validation – observed versus predicted yields for Best Subset (left) and Lasso (right).*

## (g) Prescribed folds

When the year-based split is repeated using **custom folds** (7 folds, each spanning 8 consecutive years) for cross-validation.

Lasso regression gave MSE ≈ 4.32 and $R^2 \approx 0.34$. Prescribed Folds achieved slightly better performance compared to the previous split. This indicates that consistent cross-validation improves stability in λ selection, though overall accuracy remains limited by temporal variability in the data.

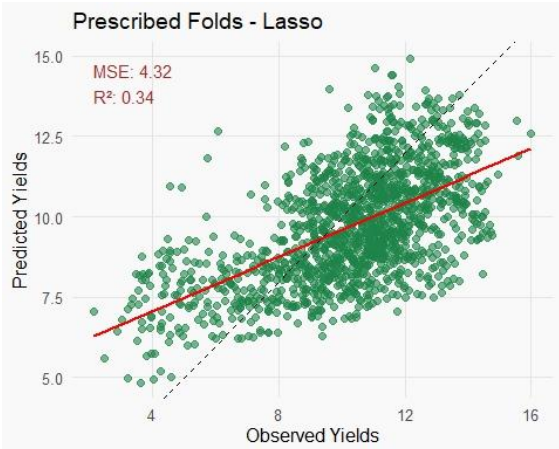


*Figure 6: Temporal split validation with prescribed folds – Observed versus predicted yield for Lasso Regression.*

# 3. Discussion

The analysis of Iowa crop yields provides valuable insights into how meteorological variables influences crop yield over a period. The dimensionality reduction analysis shows that a small number of principal components capture most of the predictor variance, suggesting that only a few dominant climatic drivers govern yield variability. Trend analysis further showed that while some locations exhibit significant changes, long-term trends are limited after multiple

testing correction, suggesting that factors beyond climate, such as technological progress also play a role.

Comparing regression strategies, both best subset and lasso regression achieved similar performance in random splits. However, year-based temporal splits produced notably weaker performance, underlining the challenge of temporal extrapolation. This strongly indicates that models trained on historical data may struggle to generalize to unseen future conditions. The prescribed cross-validation folds improved performance by enforcing a more realistic evaluation strategy. Thus climate & yield relationships might evolve, limiting the predictive power of models trained on historical data.

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# 4.Conclusion

This study applied statistical learning methods to model crop yields from meteorological predictors in the Iowa dataset. The key findings are:

**Data exploration:** The dataset contains 2875 samples, 23 predictors, and yields with mean 9.55 and standard deviation 2.84, distributed approximately bell-shaped.

**Non-i.i.d. data:** Strong spatial and temporal dependencies exist, limiting the assumption of independent samples.

**Yield trends:** Many locations show significant trends, though robustness depends on multiple-testing adjustments.

**Dimensionality (PCA):** Ten principal components capture over 75% of the variance, indicating substantial redundancy among predictors.

**Predictive performance:** Random train-test splits yield moderate accuracy ($R^2 \approx 0.56$), but temporal splits reduce predictive power ($R^2 \approx 0.30$).

**Cross-validation strategy:** Prescribed folds based on consecutive years moderately improved robustness ($R^2 \approx 0.34$).

Overall, meteorological variables alone provide useful but limited predictive accuracy for yields. This exercise underscores the importance of careful model validation strategies when applying statistical learning, especially where long-term forecasting is required.